



MUHAMMAD GHIFARI PERMATA ALWY

SOFTWARE ENGINEER | SYSTEM ANALYST

Bandung, Indonesia | alwyghifari1@gmail.com | 0851-5642-9265 | linkedin.com/in/ghifarialwy

SUMMARY

A driven and adaptable Software Engineer with a solid foundation in full-stack development, system analysis, and IT infrastructure support. Experienced in translating business needs into practical technical solutions, building web and mobile applications, and working closely with cross-functional teams to deliver results. Enthusiastic about continuous learning, eager to take on new challenges, and committed to building scalable, user-centered digital products that create real value for organizations.

PROFESSIONAL EXPERIENCE

Internship, IT Support Specialist – Project and Service Delivery 2 Division, PT Infrastruktur Telekomunikasi Indonesia

July 2025 - Sep 2025

- Developed DASH-DISM, a web-based Dashboard Analyst prototype for monitoring the dismantling of microwave antennas on the 1879 HOP project, using React.js/TypeScript for the frontend and Express.js for the backend.
- Designed and implemented role-based authentication and access control for three user roles (Project Manager, Field Coordinator, and Data Entry User).
- Built a collaborative data entry portal for daily plan (morning) and daily actual (evening) reporting, replacing manual recap via Notepad and WhatsApp groups.
- Developed an analytics dashboard featuring real-time S-Curve visualization, weekly dismantling quantity charts, and activity progress breakdown by weighted KPI.
- Implemented an automated reporting system to generate daily plan and actual reports, along with data management features including Excel import, data export, and audit logs.
- Provided hardware, software, and network troubleshooting support for internal staff, ensuring minimal downtime and smooth day-to-day operations.
- Maintained and supported the Asset Tagging and Tracking application, ensuring data accuracy and operational reliability across the division's asset records.
- Prepared technical reports and implementation documentation to support project delivery and knowledge transfer within the team.

System Analyst – Yunas Studio

Sep 2025 - July 2026

- Conducted requirements engineering through stakeholder interviews and field observation with Yunas Studio's admin, resulting in 40 Functional Requirements and 14 Non-Functional Requirements across three user roles (Customer, Admin, Superadmin).
- Modeled the existing (as-is) business process using the Functional and Object-Oriented Methodology (FOOM), producing a business Data Flow Diagram (DFD) and an Entity Relationship Diagram (ERD) covering 10 entities including Booking, Payment, Schedule, and Package.
- Designed system behavior using Object-Oriented Analysis and Design (OOAD), producing 23 Use Cases, 23 Activity Diagrams, 23 Sequence Diagrams, and a Class Diagram with 9 core classes and 5 relationship types across Customer, Admin, and Superadmin actors.
- Designed a layered system architecture (Frontend, Backend, External Services, Database) for a Progressive Web App (PWA), including integration with a Payment Gateway (Midtrans/Xendit) to automate payment verification and eliminate manual confirmation via WhatsApp.
- Authored a Software Requirements Specification (SRS) and Software Design Document (SDD), mapping all 54 requirements to design artifacts through a Requirement Traceability Matrix (RTM), achieving 100% requirement coverage.
- Facilitated stakeholder validation sessions with Yunas Studio's admin, achieving 100% approval across 10 core business process aspects, confirming the design met actual operational needs.
- Bridged communication between business stakeholders and the development team, and collaborated with UI/UX designers to validate proposed solutions against real business workflows..

Full Stack Developer (Freelance), CV. Permata Alam Perkasa

Aug 2026 - Present

- Developing and customizing an ERP system using the Odoo Framework to support end-to-end business operations of a retreaded and truck tire distribution company.
- Configuring and managing the Inventory module to track stock levels, warehouse locations, and product movement.
- Setting up the Sales and Purchase modules to manage product distribution, incoming and outgoing goods, and vendor/customer transactions.
- Implementing the Accounting module to support financial recording and reporting aligned with business processes.
- Configuring the Employees/Attendance module to manage employee data and daily attendance tracking.
- Managing the Product module, including product data, categories, and pricing structures within the ERP system.

PROJECTS

Full Stack Developer – StudyIT & StudyIT Mobile

Sep 2023 - June 2024

- Developed and maintained web and mobile application features using React.js and Flutter.
- Built responsive user interfaces for both web and mobile platforms.
- Implemented MVC (Model-View-Controller) architecture on the web backend using Laravel, separating Eloquent models (Course, Instructor, Enrollment, Discussion, Account, etc.) from role-based Blade views (Admin and Regular User) and dedicated controller classes.
- Built a RESTful API service with Express.js following an OOP-based MVC pattern, with dedicated model and controller classes per resource (Course, Account, Instructor, Enrollment, Discussion, Task, Reply, Item, Module).
- Implemented authentication and security features, including JWT-based auth, two-factor authentication (OTP), and file upload handling (Multer).
- Collaborated on learning platform architecture and content delivery.

Restaurant Order Management System – WinForms Desktop App

June 2024

Back-End & Front-End Developer | Academic Project (Konstruksi Perangkat Lunak)

- Built a desktop-based restaurant order management system using C# and Windows Forms (.NET Framework), following the MVC (Model-View-Controller) architecture to separate business logic, UI, and data layers.
- Developed the Login and Homepage modules, implementing input validation and secure coding practices (null/empty checks, exception handling) to prevent invalid data submission and application crashes.
- Implemented JSON-based data persistence using System.Text.Json for reading and writing user account data.
- Modeled order processing logic using a table-driven state machine (Pending → Processing → Completed/Cancelled) with generic classes and Design by Contract principles.

ShelfLink – Digital Library Management System | Full-Stack Developer

Sep 2025 - June 2026

Academic Startup Project

- Developed a web-based library management system to digitalize book cataloging, borrowing, returning, and review processes.
- Designed and implemented both frontend and backend features using Laravel and MySQL, following an MVC architecture with dedicated model and controller classes (Book, Borrowing, BookReview, User, Catalog, Return, Statistics).
- Integrated an AI-based image analysis service to automatically detect book damage from return photos, generating a confidence score and a suggested fine.
- Built an automated fine calculation system combining overdue penalties and AI-suggested damage fines, with automatic overdue status detection.
- Built CRUD modules for book management, user management, borrowing, and return transactions.
- Developed authentication and role-based access control for administrators and library users.
- Implemented automated email notifications (borrow confirmation, overdue reminders, return receipts) using Laravel Mail.
- Optimized database structure and RESTful application workflows to improve system performance and maintainability.
- Set up a CI/CD pipeline using GitHub Actions to run automated PHPUnit tests, integrated with SonarCloud for continuous code quality analysis.
- Participated in MVP development, functional testing, and usability testing, contributing to product improvements based on user feedback.
- Collaborated with the project team using Git throughout the software development lifecycle.

Full Stack Developer | Aplikasi Pendaftaran Beasiswa Pelatihan (Recruitment PT. Bentang Inspirasi Teknologi)

September 2026

- Designed and built a 6-service microservices architecture (API Gateway, RBAC, Master Data, Transaction, Document, Frontend) using Node.js 20, TypeScript, Express, Prisma, and React 18, orchestrated via Docker Compose.
- Implemented asymmetric JWT authentication (RS256) with private/public key separation between the auth service and gateway, argon2id password hashing, and refresh-token rotation with reuse detection.
- Enforced a private-network security model where all backend services and databases are isolated from the host (Docker internal network), accessible only through a single API Gateway with rate limiting, CORS whitelisting, and Helmet security headers.
- Built a secure document upload pipeline with magic-byte file-type validation, ClamAV antivirus scanning, UUID-based storage, and SHA-256 checksums to prevent file-spoofing and unauthorized access.
- Designed a database schema across MySQL (RBAC, Master, Transaction) and PostgreSQL (Document, using JSONB metadata), covering 14 tables with a state-machine-driven registration workflow (draft → submitted → verified → interviewed → passed).
- Wrote unit tests (Vitest) covering authentication, token rotation, and file-type validation, and produced full technical documentation (ERD, OpenAPI spec, security audit, architecture decision records).

EDUCATION

Bachelor of Software Engineering (S1 Rekayasa Perangkat Lunak)
School of Computing, Telkom University
GPA: 3.38/4.00

2022 - 2026

SKILLS

System Architecture, Mobile Development, Software System Analysis, Data Analysis, Requirement Analysis, Artificial Intelligence System Development, DevOps, Front-End Development, Back-End Development, Teamwork, Project Management

ADDITIONAL INFORMATION

- Languages: Bahasa, English
- Certifications: Hak Kekayaan Intelektual (HKI) – Sistem Booking pada Foto Studio berbasis PWA (Progressive Web App)
- IBM Full-Stack Software Developer (Coursera), 2025
- Meta Full-Stack Software Developer (Coursera), 2025

ORGANIZATIONS

- Professional Development Staff - BEM KEMA Telkom University,
 - Staff of External Division - HMRPL (Himpunan Mahasiswa RPL),
 - Chairman - PERMALA (Perhimpunan Mahasiswa Lampung) Telkom University,
- 2022 - 2023**
2023 - 2024
2024 - 2025



Scan to my
Portofolio